This assignment will not be graded and is only for practice.

Level 1

1) Choose the set of correct options. ***1 point***

- [ ] The set $\{(1,2,3),(4,5,6),(7,8,9)\}$ is linearly dependent.
- [ ] The set $\{(1,2,3),(4,5,6),(5,7,9)\}$ is linearly independent.
- [ ] The set $\{(1,2,3),(0,5,6),(0,0,9)\}$ is linearly independent.
- [ ] The set $\{(1,2,3),(0,0,6),(0,0,9)\}$ is linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(1,2,3),(4,5,6),(7,8,9)\}$ is linearly dependent.
The set $\{(1,2,3),(0,5,6),(0,0,9)\}$ is linearly independent.

2) If $ad - bc = 0$ then which of the following options are true? ***1 point***

- [ ] The set $\{(a,b),(c,d)\}$ is linearly dependent in $\mathbb{R}^2$.
- [ ] The set $\{(a,c),(b,d)\}$ is linearly dependent in $\mathbb{R}^2$.
- [ ] The set $\{(1,0,0),(0,a,b),(0,c,d)\}$ is linearly dependent in $\mathbb{R}^3$.
- [ ] The set $\{(1,0,0),(0,a,c),(0,b,d)\}$ is linearly dependent in $\mathbb{R}^3$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(a,b),(c,d)\}$ is linearly dependent in $\mathbb{R}^2$.
The set $\{(a,c),(b,d)\}$ is linearly dependent in $\mathbb{R}^2$.
The set $\{(1,0,0),(0,a,b),(0,c,d)\}$ is linearly dependent in $\mathbb{R}^3$.
The set $\{(1,0,0),(0,a,c),(0,b,d)\}$ is linearly dependent in $\mathbb{R}^3$.

3) For $c \in \mathbb{Z}$, consider the set of three vectors $S = \{(1,c,0),(c,0,c),(2c,3c,4c)\}$ in $\mathbb{R}^3$ with usual addition and scalar multiplication. Find the value of c such that the given set is linearly dependent.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

4) Assume the set $\{(a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3)\}$ is given to be linearly independent and consider the following system of linear equations ***1 point***

$$a_1x + b_1y + c_1z = 1$$
$$a_2x + b_2y + c_2z = 2$$
$$a_3x + b_2y + c_3z = 3.$$

If $AX = b$ is matrix representation of the above system of linear equations, where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$,

then which of the following option(s) is(are) true?

- [ ] The system of linear equations has a unique solution.
- [ ] The system of linear equations has no solution.
- [ ] The system of linear equations have infinitely many solutions.
- [ ] The coefficient matrix A is invertible.
- [ ] The coefficient matrix A is not invertible.
- [ ] Any one of the above options cannot be concluded using the given information.

No, the answer is incorrect.
Score: 0

Accepted Answers:

The system of linear equations has a unique solution.
The coefficient matrix A is invertible.

5) ***1 point***

Assume the set $\{(a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3)\}$ is given to be linearly dependent and consider the following system of linear equations

$$\begin{aligned} a_1x + b_1y + c_1z &= 1 \\ a_2x + b_2y + c_2z &= 2 \\ a_3x + b_2y + c_3z &= 3. \end{aligned}$$

If $AX = b$ is matrix representation of the above system of linear equations, where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$,

then which of the following option(s) is(are) true?

- [ ] Either the system of linear equations has a unique solution or no solution.
- [ ] Either the system of linear equations has no solution or infinitely many solutions.
- [ ] Either the system of linear equations have infinitely many solutions or a unique solution.
- [ ] The system of linear equations has a unique solution.
- [ ] The coefficient matrix A is invertible.
- [ ] The coefficient matrix A is not invertible.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Either the system of linear equations has no solution or infinitely many solutions.
The coefficient matrix A is not invertible.

6) If S is a linearly independent set in vector space $\mathbb{R}^3$ with respect to usual addition and scalar multiplication, then find the maximum possible cardinality of S.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Level 2

7) Let the set $\{(a, b), (c, d)\}$ be a linearly independent subset of $\mathbb{R}^2$. Choose the set of correct options. ***1 point***

- [ ] $\{(a, b, 0), (c, d, 0)\}$ must be a linearly independent subset of $\mathbb{R}^3$.
- [ ] $\{(a, b, 0), (c, d, 0), (0, 0, 1)\}$ must be a linearly independent subset of $\mathbb{R}^3$.
- [ ] $\{(a, b, 0), (c, d, 0), (1, 0, 0)\}$ must be a linearly independent subset of $\mathbb{R}^3$.
- [ ] $\{(a, b, 0), (c, d, 0), (0, 1, 0)\}$ must be a linearly independent subset of $\mathbb{R}^3$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(a, b, 0), (c, d, 0)\}$ must be a linearly independent subset of $\mathbb{R}^3$.
$\{(a, b, 0), (c, d, 0), (0, 0, 1)\}$ must be a linearly independent subset of $\mathbb{R}^3$.

8) Each column of an $n \times n$ matrix A can be thought of as vectors in $\mathbb{R}^n$. If a set S consists of all the column vectors of A, then which of the following is(are) true? ***1 point***

- [ ] S is always linearly independent.
- [ ] S is linearly independent if $det(A) = 0$.
- [ ] S is linearly independent if $det(A) = 0$.
- [ ] If $det(A) = 1$, then S is linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

S is linearly independent if $det(A) = 0$.
If $det(A) = 1$, then S is linearly independent.

9) ***1 point***

Suppose it is known that the three vectors $v_1 = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, v_2 = \begin{bmatrix} d \\ e \\ f \end{bmatrix}, v_3 = \begin{bmatrix} g \\ h \\ i \end{bmatrix}$ are linearly dependent in $\mathbb{R}^3$. Which of the following is correct?

- [ ] Some linear combination of a, b, c must be 0.
- [ ] Some linear combination of a, d, g must be 0.
- [ ] Some linear combination of d, e, f must be 0.
- [ ] Some linear combination of c, f, i must be 0.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Some linear combination of a, b, c must be 0.
Some linear combination of a, d, g must be 0.
Some linear combination of d, e, f must be 0.
Some linear combination of c, f, i must be 0.

10) If vectors $(2x, 4, -6), (-6, -3x, 2x), (-1, x-3, x+2)$ from the vector space $\mathbb{R}^3$ with usual addition and scalar multiplication, are linearly dependent and $x \in [2, 5]$, then find the value of x.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Your score is: 0/10